# Lecture 1: Introduction

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**MSA220/MVE441** Statistical Learning for Big Data

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CHALMERS UNIVERSITY OF TECHNOLOGY | UNIVERSITY OF GOTHENBURG

## The Data Deluge

- Data is getting cheap! (sometimes)
- Massive data collections across all fields! Molecular biology, health care, banking, streaming, marketing, citizen science, climate modeling, imaging, data feeds like twitter, ...
- What do we do with all this data?

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- but, if we collect tons of information about objects just because we can - do we risk drowning in uninformative data?
- many methods perform poorly if we overwhelm them with high-dimensional data
- is there a selection bias when we collect high-dimensional data? - which features of objects are easy to obtain?

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- Big n can lead to computational problems, can be more difficult to handle at the data exploration stage, difficult to visualize and assess model adequacy
- Big n statistics: with a big enough n everything becomes significant - think carefully about what the analysis goals are

# Big Data = Big problems? It's a huge topic in science!

Lot's of research! Companies need the competence!

- Logistics, transport, banking, risk analysis, automated detection, image and video processing, molecular and systems biology
- and more...
- Methodology research: dimension reduction and visualization, computing solutions, feature selection/interpretability, scalable algorithms, ....

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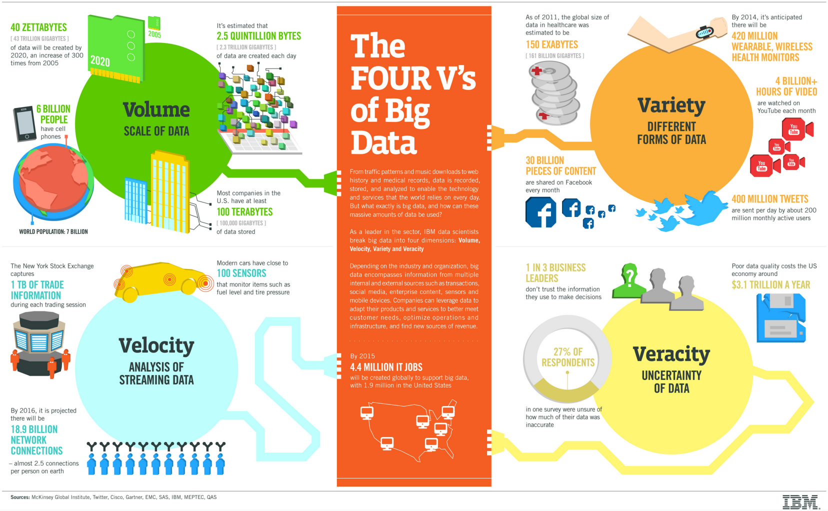
Is this all?

# The Four Vs of Big Data




http://web.archive.org/web/20210506042232/https://www.ibmbigdatahub.com/infographic/four-vs-big-data

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Focus is on the last three in this course.

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- Exploration and visualisation of Big Data can already require statistics
- **Probability of extreme values:** Unlikely results become much more likely with an increase in $n$
- **Curse of dimensionality:** Lot's of space between data points in high-dimensional space

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- In fact: **with Big Data you may encounter selection bias, mislabeled observations, imbalance of data, noisy data, spurious correlations, ....**
- Before analysis, spend time with the people collecting the data to try to understand how, why, think about sources for bias, questions that were *not* asked, .... **Taking time to understand the data will save you a lot of headache later on!**

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  - Significance vs Importance
  - Overfitting
  - Computational burden
  - but also blessing (e.g. SGD and ensembles, ”data hungry” methods like NN, rare-events detection, noise correction)

## PAUSE!!!!

A bit of recap.....

If what comes next feels unfamiliar - spend a bit of time reviewing basic stat concepts

1. Statistics terminology recap
2. Basics of statistical learning
3. Setting the stage for the course

## Basics about random variables

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- **Probability mass function (pmf)** $p(k)$ for a discrete variable

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- Where will we see this? **Classification**, where $\theta$ denotes the probability of class 1 and can be observation specific, $\theta_i$, and depend on features of this observation, $\theta(x_i)$

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# Basics about random variables

- We will consider **discrete** and **continuous** random quantities
- **Probability density function (pdf)** $p(\mathbf{x})$ for a continuous variables

  **Example:** Multivariate normal distribution with mean vector $\boldsymbol{\mu} \in \mathbb{R}^p$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$

$$p(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

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# Two important rules (and a consequence)

## Marginalisation

For a joint density $p(x, y)$ it holds that

$$p(x) = \sum_y p(x, y) \quad \text{or} \quad p(x) = \int p(x, y)\,\mathrm{d}y$$

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Both rules together imply **Bayes' law**

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

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- Bayes law: the basis for building classification rules by updating marginal population frequencies once we observe observation specific features.

## Expectation and variance

Expectations and variance depend on an underlying pdf/pmf.

**Notation:**

- $\mathbb{E}_{p(x)}[f(x)] = \int f(x) p(x) \, \mathrm{d}x$
- $\mathrm{Var}_{p(x)}[f(x)] = \mathbb{E}_{p(x)}\left[\left(f(x) - \mathbb{E}_{p(x)}[f(x)]\right)^2\right]$

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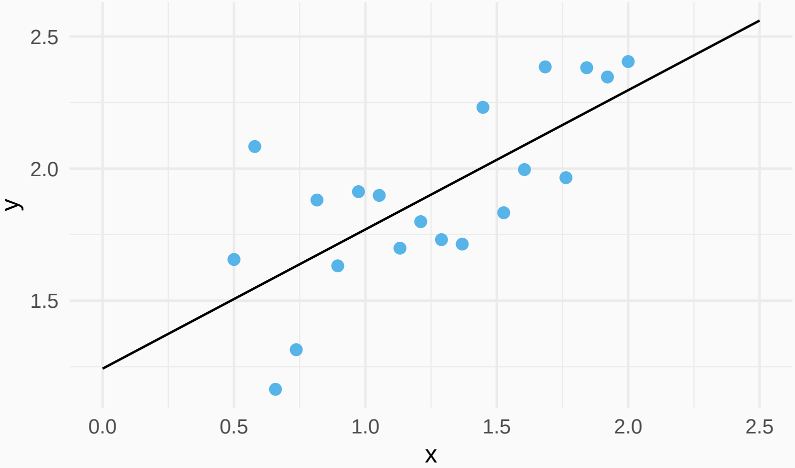
- **Loss function:** Quantifies the discrepancy between observed data and predictions

# Linear regression - An old friend



- **Data:** Training data consists of independent pairs

$$(y_i, \mathbf{x}_i), \quad i = 1, \ldots, n$$

Observed response $y_i \in \mathbb{R}$ for predictors $\mathbf{x}_i \in \mathbb{R}^p$

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$$L(y, \hat{y}) = (y - \hat{y})^2$$

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- What can happen if these are violated?
- What, if anything, can we do to handle violations?

## Statistical decision theory for regression (I)

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$$J(f) = \mathbb{E}_{p(\mathbf{x},y)}\left[L(y, f(\mathbf{x}))\right] = \mathbb{E}_{p(\mathbf{x})}\left[\mathbb{E}_{p(y|\mathbf{x})}\left[L(y, f(\mathbf{x}))\right]\right]$$

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- Define 'best' by:

$$\widehat{f} = \arg\min_f J(f)$$

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$\mathbb{E}_{p(y|\mathbf{x})}[y]$ : "Average of $y$ in vertical slice defined by $x$"
f(x): Our model

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Can we determine $\widehat{f}$? Focus on inner expectation

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\mathbb{E}_{p(y|\mathbf{x})}\left[(y - f(\mathbf{x}))^2\right] &= \int (y - \mathbb{E}_{p(y|\mathbf{x})}[y] + \mathbb{E}_{p(y|\mathbf{x})}[y] - f(\mathbf{x}))^2 p(y|\mathbf{x})\,\mathrm{d}y \\
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&= \mathrm{Var}_{p(y|\mathbf{x})}[y] + (\mathbb{E}_{p(y|\mathbf{x})}[y] - f(\mathbf{x}))^2
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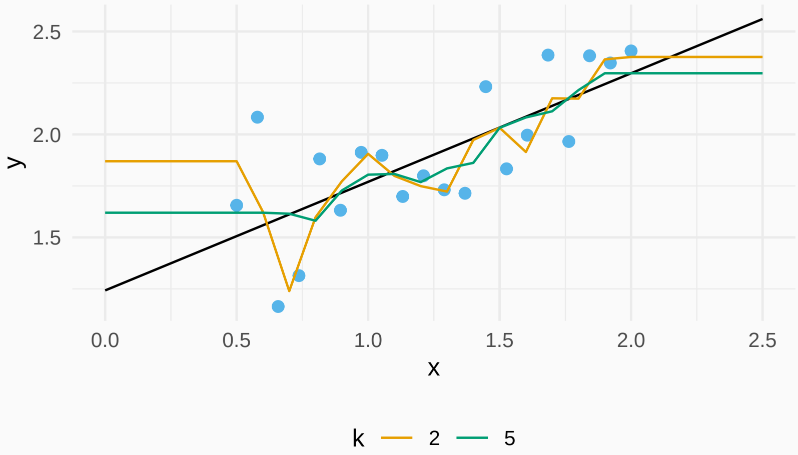
- Probably more realistic to look for the $k$ closest neighbours of $\mathbf{x}$ in the training data $N_k(\mathbf{x}) = \{\mathbf{x}_{i_1}, \ldots, \mathbf{x}_{i_k}\}$. Then

$$\mathbb{E}_{p(y|\mathbf{x})}[y] \approx \frac{1}{k} \sum_{\mathbf{x}_{i_l} \in N_k(\mathbf{x})} y_{i_l}$$

# Average of $k$ neighbours



# Back to linear regression

Linear regression is a **model-based approach** and assumes that the dependence of $y$ on $\mathbf{x}$ can be written as a weighted sum, i.e.

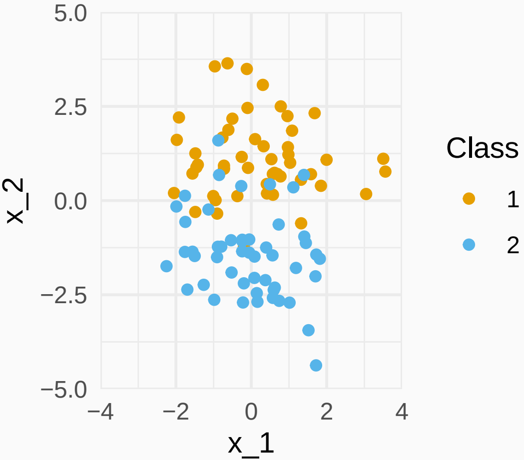
$$y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon$$

where $\varepsilon \sim N(0, \sigma^2)$. The mean of $y$ given $\mathbf{x}$ is therefore

$$\mathbb{E}_{p(y|x)}[y] = \mathbf{x}^\top \boldsymbol{\beta}.$$

Note that in practice this equality will only hold approximately.

# A simple example of classification



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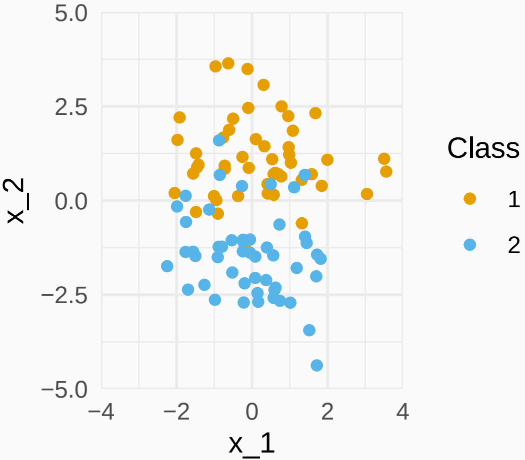


How do we classify a pair of new coordinates $\mathbf{x} = (x_1, x_2)$?

# $k$-nearest neighbour classifier (kNN)

- Find the $k$ predictors

$$N_k(\mathbf{x}) = \{\mathbf{x}_{i_1}, \dots, \mathbf{x}_{i_k}\}$$

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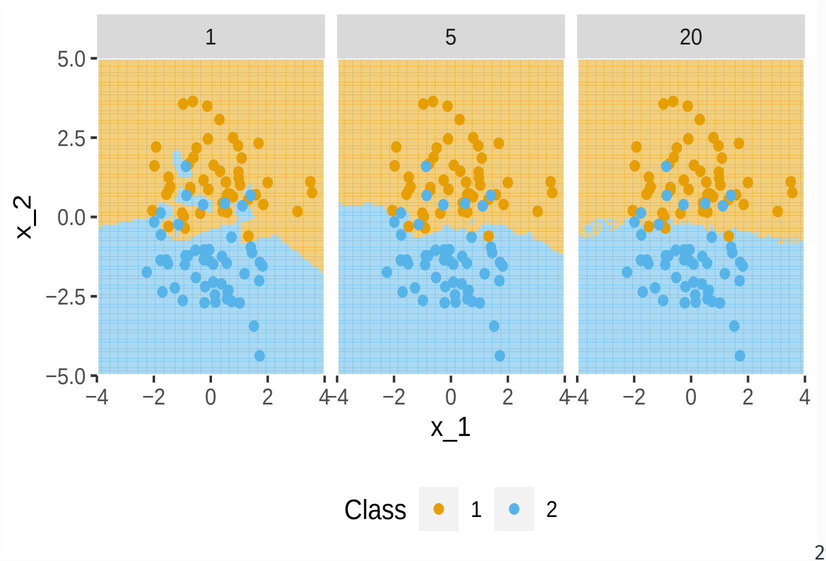
- **Majority vote:** Assign $\mathbf{x}$ to the class that most predictors in $N_k(\mathbf{x})$ belong to (highest frequency)

# kNN and its decision boundaries



**Classification**

Learn a rule $c(\mathbf{x})$ from data which maps observed features $\mathbf{x}$ to classes $\{1, \ldots, K\}$.

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**What is a suitable loss?**

# Statistical decision theory for classification

- **0-1 misclassification loss:** Let $i$ be the actual class of an object and $c(\mathbf{x})$ is a rule that returns the class for the variable(s) $\mathbf{x}$, then

$$L(i, c(\mathbf{x})) = \begin{cases} 0 & i = c(\mathbf{x}), \\ 1 & i \neq c(\mathbf{x}) \end{cases} = \mathbb{1}(i \neq c(\mathbf{x}))$$

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- Minimizing expected prediction error leads to the rule

$$\hat{c}(\mathbf{x}) = \arg\max_{1 \le i \le K} p(i|\mathbf{x})$$

This is called **Bayes' rule**.

## Deriving Bayes' rule

Again, focus on inner expectation

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Minimal for $\hat{c}(\mathbf{x}) = \arg\max_{1 \leq i \leq K} p(i|\mathbf{x})$

- kNN solves the classification problem by approximating $p(i|\mathbf{x})$ with the frequency of class $i$ among the $k$ closest neighbours of $\mathbf{x}$.
- Given data $(i_l, \mathbf{x}_l)$ for $l = 1, \ldots, n$ it holds that

$$\hat{c}(\mathbf{x}) = \underset{1 \leq i \leq K}{\arg\max} \frac{1}{k} \sum_{\mathbf{x}_l \in N_k(\mathbf{x})} \mathbb{1}(i_l = i)$$

# A note on kNN

There are two choices to make when implementing a kNN method

1. The metric to determine a neighbourhood
    - e.g. Euclidean/$\ell_2$ norm, Manhattan/$\ell_1$ norm, max norm, …
2. The number of neighbours, i.e. $k$

The choice of metric changes the underlying local model of the method while $k$ determines the size of this local model.

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- **Point being that the same principles underpin these methods and the key will be how to train, tune and validate them.**

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## Take-home message

- Big Data is complex and is multi-faceted
- Regression and classification can be formulated in the framework of Statistical Learning
- In both cases, focus is on prediction
- Next class: Dimension reduction, Logistic regression and Bias-Variance trade-off